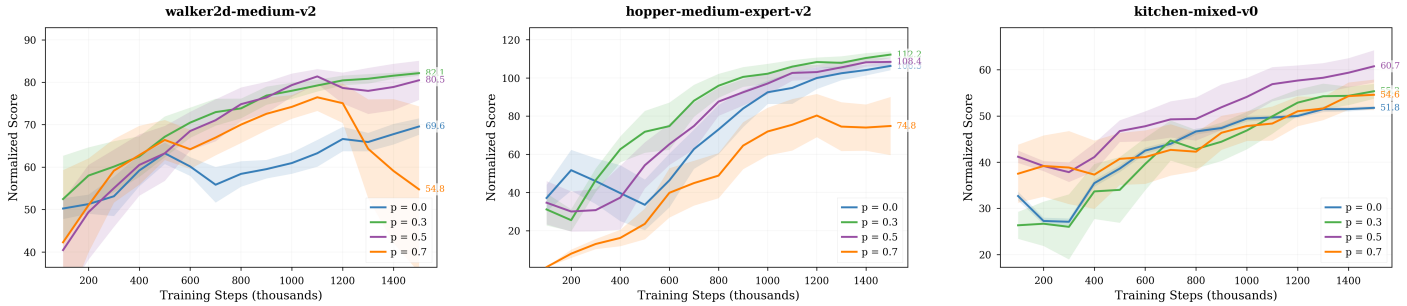

Figure 1: Sensitivity analysis of the stochasticity parameter p on representative D4RL tasks (`walker2d-medium`, `hopper-medium-expert`, and `kitchen-mixed-v0`), complementing the `halfcheetah-medium-replay-v2` results from our initial submission. We selected these specific environments to sample diverse qualities across both locomotion and manipulation domains. Moderate values ($p \in \{0.3, 0.5\}$) consistently yield the highest normalized scores, while performance degrades at $p = 0.7$ (extreme values, e.g., $p \geq 0.9$, diverge and are omitted for visual clarity). Solid lines show the mean across 3 random seeds with EMA smoothing (weight = 0.6); shaded regions denote ± 1 SEM.

Table 1: Hyperparameters for the base Gaussian ISEP across D4RL tasks. The table details the interpolation parameter (p), expectile (τ), and the temperature (β).

Task	p	τ **(expectile)**	β **(temperature)**
halfcheetah-medium-v2	0.5	0.8	5.0
hopper-medium-v2	0.5	0.8	3.0
walker2d-medium-v2	0.3	0.8	5.0
halfcheetah-medium-replay-v2	0.5	0.8	5.0
hopper-medium-replay-v2	0.3	0.8	3.0
walker2d-medium-replay-v2	0.5	0.8	5.0
halfcheetah-medium-expert-v2	0.3	0.8	3.0
hopper-medium-expert-v2	0.3	0.8	3.0
walker2d-medium-expert-v2	0.3	0.8	3.0
pen-human-v1	0.2	0.7	0.5
pen-cloned-v1	0.5	0.7	0.5
kitchen-complete-v0	0.5	0.7	0.05
kitchen-partial-v0	0.2	0.8	0.1
kitchen-mixed-v0	0.3	0.8	0.1

Table 2: Hyperparameters for the ISEP-FM variant across D4RL tasks. The **guidance_weight** (w) denotes the classifier-free guidance weight applied during inference. Analogous to the temperature (β) in base ISEP, this guidance weight controls the degree of policy improvement.

Task	p	τ **(expectile)**	w **(guidance_weight)**
halfcheetah-medium-v2	0.3	0.7	2.0
hopper-medium-v2	0.2	0.7	1.0
walker2d-medium-v2	0.3	0.8	2.0
halfcheetah-medium-replay-v2	0.3	0.8	2.0
hopper-medium-replay-v2	0.5	0.8	3.0
walker2d-medium-replay-v2	0.3	0.8	3.0
halfcheetah-medium-expert-v2	0.3	0.8	3.0
hopper-medium-expert-v2	0.2	0.7	3.0
walker2d-medium-expert-v2	0.3	0.7	2.0
pen-human-v1	0.2	0.7	3.0
pen-cloned-v1	0.2	0.7	1.0
kitchen-complete-v0	0.2	0.8	0.5
kitchen-partial-v0	0.2	0.8	1.0
kitchen-mixed-v0	0.2	0.8	0.2

Table 3: Normalized scores on D4RL MuJoCo locomotion tasks comparing our proposed ISEP and ISEP-FM against the Decision Transformer (DT) baseline. Best overall scores are highlighted in **bold**. Our results (ISEP and ISEP-FM) are averaged over 5 random seeds, while DT results are reported directly from the original publication.

Environment	**DT**	**ISEP**	**ISEP-FM**
halfcheetah-med	42.6	49.3 ± 0.2	**59.9 ± 0.3**
hopper-med	67.6	79.1±3.5	**101.3 ± 2.1**
walker2d-med	74.0	82.1±0.4	**90.4 ± 1.1**
halfcheetah-med-rep	36.6	46.3±0.1	**54.6 ± 0.5**
hopper-med-rep	82.7	102.7±0.6	**104.5 ± 0.8**
walker2d-med-rep	66.6	82.0±3.6	92.7 ± 1.5
halfcheetach-med-exp	86.8	92.4±0.3	**95.7 ± 0.2**
hopper-med-exp	107.6	112.0±0.2	**112.3 ± 0.3**
walker2d-med-exp	108.1	110.2±0.5	**112.5 ± 0.4**
locomotion-v2 total	74.7	84.0	**91.5**